THE DIAGONAL OF A FORMAL ALGEBRAIC SPACE IS REPRESENTABLE, A MONOMORPHISM, LOCALLY QUASI-FINITE, LOCALLY OF FINITE TYPE AND SEPARATED

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays a chain of identifications of fibre products and a composition of morphisms; both lie inside the counted span and are reproduced verbatim.

Editorial note. All references of the proof point outside the counted span and are carried as navigation stubs; no same-chapter context span is required and none was added.

Editorial note. The counted spans carry 12 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The presentation of a formal algebraic space by affine formal algebraic spaces. Same chapter, Definition definition-formal-algebraic-space, invoked at proof lines 2675-2676. Supplies the covering whose base change produces the jointly surjective family.
- The diagonal of an affine formal algebraic space, and representability of the members of a presentation. Same chapter, Lemma lemma-diagonal-affine-formal-algebraic-space at proof lines 2681-2682 and 2701-2704, and Lemma lemma-presentation-representable at proof lines 2682-2683. Give that the members of the family are etale over the fibre product with sources that are schemes, and that the relevant morphisms are closed immersions.
- The bootstrap theorem producing an algebraic space from a jointly surjective family. Bootstrap, Theorem bootstrap-theorem-final-bootstrap, invoked at proof lines 2686-2688. Makes the fibre product an algebraic space.
- A locally quasi-finite separated algebraic space over a scheme is a scheme, and properties of representable transformations may be checked on schemes. Morphisms of Spaces, Proposition spaces-morphisms-proposition-locally-quasi-finite-separated-over-scheme at proof lines 2707-2709, and Spaces,

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Lemmas spaces-lemma-representable-diagonal, spaces-lemma-transformation-diagonal-properties and spaces-lemma-representable-transformations-property-implication at proof lines 2709-2712 and 2705-2708. Conclude representability and then transfer the remaining four properties from the scheme case.

- Locality of local quasi-finiteness on the source, and the corresponding statements for schemes. Descent on Spaces, Lemma spaces-descent-lemma-locally-quasi-finite-etale-local-source at proof lines 2701-2705, Schemes, Lemma schemes-lemma-monomorphism-separated at proof lines 2709-2711, and Morphisms, Section morphisms-section-quasi-finite at proof lines 2711-2712. Give that local quasi-finiteness can be checked after the etale base change, that a monomorphism of schemes is separated, and that a locally quasi-finite morphism of schemes is locally of finite type.

Source reference. Original theorem 04S6: sources/stacks-chow/tags.txt, source lines 6209–6209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AI8: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2010–2010. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIG: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2381–2381. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 01TC: sources/stacks-chow/tags.txt, source lines 2369–2369. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01L4: sources/stacks-chow/tags.txt, source lines 2073–2073. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06F5: sources/stacks-chow/tags.txt, source lines 8239–8239. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 025W: sources/stacks-chow/tags.txt, source lines 2829–2829. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02YO: sources/stacks-chow/tags.txt, source lines 3883–3883. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0CB7: sources/stacks-chow/tags.txt, source lines 15462–15462. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 03XX: sources/stacks-chow/tags.txt, source lines 5164–5164. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. If X is a formal algebraic space over S, then the diagonal morphism $\Delta : X \to X \times_S X$ is representable, a monomorphism, locally quasi-finite, locally of finite type, and separated.*

Proof. Suppose given $U \to X$ and $V \to X$ with U, V schemes over S. Then $U \times_X V$ is a sheaf. Choose $\{X_i \to X\}$ as in Definition 0.1. For every i the morphism

$$(U \times_X X_i) \times_{X_i} (V \times_X X_i) = (U \times_X V) \times_X X_i \to U \times_X V$$

is representable and étale as a base change of $X_i \to X$ and its source is a scheme (use Lemmas 0AI8 and 0AIG). These maps are jointly surjective hence $U \times_X V$ is an algebraic space by Bootstrap, Theorem 04S6. The morphism $U \times_X V \to U \times_S V$ is a monomorphism. It is also locally quasi-finite, because on precomposing with the morphism displayed above we obtain the composition

$$(U \times_X X_i) \times_{X_i} (V \times_X X_i) \to (U \times_X X_i) \times_S (V \times_X X_i) \to U \times_S V$$

which is locally quasi-finite as a composition of a closed immersion (Lemma 0AI8) and an étale morphism, see Descent on Spaces, Lemma 06F5. Hence we conclude that $U \times_X V$ is a scheme by Morphisms of Spaces, Proposition 03XX. Thus Δ is representable, see Spaces, Lemma 025W.

In fact, since we've shown above that the morphisms of schemes $U \times_X V \to U \times_S V$ are always monomorphisms and locally quasi-finite we conclude that $\Delta : X \to X \times_S X$ is a monomorphism and locally quasi-finite, see Spaces, Lemma 0CB7. Then we can use the principle of Spaces, Lemma 02YO to see that Δ is separated and locally of finite type. Namely, a monomorphism of schemes is separated (Schemes, Lemma 01L4) and a locally quasi-finite morphism of schemes is locally of finite type (follows from the definition in Morphisms, Section 01TC). □